

Adaptive Law-Based Transformation *altx*

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Data and Compute Intensive Sciences Research Group

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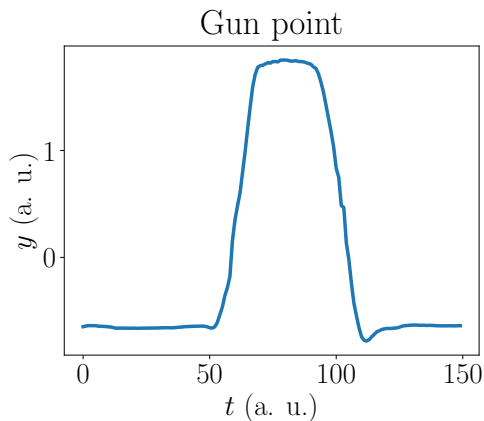
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Time series data



Time series is some function of time, for example:

$$x = [1, 2, 5, -1, 3]. \quad (1)$$

Dataset

Consider the following dataset:

- ▶ Train instance from class **a**: $a = [1, 1, 2, 3, 5]$.
- ▶ Train instance from class **b**: $b = [2, 1, 5, 7, 17]$.
- ▶ Test instance from an unknown class: $x = [2, 3, 5, 8, 13]$.

We now apply *altx* to classify the test instance, i.e., determine to which class x belongs.

Goal of *altx*

Evaluate how much the data belongs to one class (**closer to 0, more similar!**)

Data	Class a	Class b
$x_0^{(a)}$	0.2	3
$x_1^{(a)}$	0.03	5
$x_2^{(a)}$	0.1	7
$x_3^{(b)}$	4	0.1
$x_4^{(b)}$	6	0.15
$x_5^{(b)}$	2	0.3
$x^?$	0.00035	0.029

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Step 1: Extracting the Laws

We define parameters $r = 3$, $l = 2$, $k = 1$. (Their meaning will be discussed later.)

From $a = [1, 1, 2, 3, 5]$ we extract the following segments of length $r = 3$:

$$[1, 1, 2], [1, 2, 3], [2, 3, 5]. \quad (2)$$

Then we embed them to $l \times l = 2 \times 2$ matrices:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}. \quad (3)$$

From those, the laws¹ vectors which gives close to 0 when multiplying the embedded matrices, are:

$$\begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}, \quad \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}, \quad \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}. \quad (4)$$

¹eigenvectors corresponding to the smallest absolute eigenvalues

Step 1: Extracting the Laws (continued)

We combine the results into one matrix and repeat the process for all training instances (separately for each class).

From class **a**, we obtain laws:

$$L_a = \begin{bmatrix} -0.8507 & -0.8507 & -0.8507 \\ 0.5257 & 0.5257 & 0.5257 \end{bmatrix}. \quad (5)$$

From class **b**, we obtain laws:

$$L_b = \begin{bmatrix} -0.9571 & -0.8702 & -0.9085 \\ 0.2898 & 0.4927 & 0.4179 \end{bmatrix}. \quad (6)$$

Step 2: Embedding the Test Instance

The test instance $x = [2, 3, 5, 8, 13]$ is embedded as:

$$E = \begin{bmatrix} 2 & 3 \\ 3 & 5 \\ 5 & 8 \\ 8 & 13 \end{bmatrix}. \quad (7)$$

Step 3: Projection by Laws

Multiplying with class **a** laws:

$$F_a = EL_a = \begin{bmatrix} -0.0344 & -0.0344 & -0.0344 \\ 0.0132 & 0.0132 & 0.0132 \\ -0.0050 & -0.0050 & -0.0050 \\ 0.0019 & 0.0019 & 0.0019 \end{bmatrix}. \quad (8)$$

Multiplying with class **b** laws:

$$F_b = EL_b = \begin{bmatrix} -0.2898 & -0.0727 & -0.1563 \\ -0.2439 & -0.0252 & -0.1091 \\ -0.2615 & -0.0434 & -0.1272 \\ -0.2548 & -0.0365 & -0.1203 \end{bmatrix}. \quad (9)$$

Step 4: Feature Calculation

Using the mean of all squared values:

- ▶ Class **a**: $\text{mean} = 0.00035$.
- ▶ Class **b**: $\text{mean} = 0.029$.

Conclusion: The test instance is closer to class **a**.

Exercise

Using *altx_intro.ipynb* notebook calculate to which class does

$$x = [3, -1, 5, 3, 13] \tag{10}$$

belongs.

Exercise – solution

Using *altx_intro.ipynb* notebook calculate to which class does

$$x = [3, -1, 5, 3, 13] \tag{11}$$

belongs.

Class a score: 0.43, class b score: 0.42 $\Rightarrow$ it belongs to **class b**.

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Meaning of Parameters r, l, k

- ▶ r : Length of the original sequence window used for feature extraction.
- ▶ l : Embedding size, defines a symmetric $l \times l$ matrix.
- ▶ $2l - 1$: Number of equally spaced elements sampled from the window of length r .
- ▶ k : Step size (stride) used for shifting the window over the time series.

Example r, l, k values

Consider the following instance:

$$x' = [2, 1, 5, 7, 17, 31, 65, 127, 257, 511].$$

Let $r = 5, l = 2$, and $k = 1$. Then $2l - 1 = 3$ and

$$x'_1 = [\boxed{2}, 1, \boxed{5}, 7, \boxed{17}], 31, 65, 127, 257, 511].$$

$$x'_2 = [2, \boxed{1}, 5, \boxed{7}, 17, \boxed{31}], 65, 127, 257, 511].$$

$$x'_3 = [2, 1, \boxed{5}, 7, \boxed{17}, 31, \boxed{65}], 127, 257, 511].$$

$$x'_4 = [2, 1, 5, \boxed{7}, 17, \boxed{31}, 65, \boxed{127}], 257, 511].$$

$$x'_5 = [2, 1, 5, 7, \boxed{17}, 31, \boxed{65}, 127, \boxed{257}], 511].$$

$$x'_6 = [2, 1, 5, 7, 17, \boxed{31}, 65, \boxed{127}, 257, \boxed{511}].$$

Example r, l, k values

Consider the following instance:

$$x'' = [2, 1, 5, 7, 17, 31, 65, 127, 257, 511].$$

Let $r = 7, l = 2$, and $k = 2$. Then $2l - 1 = 3$ and

$$x_1'' = [\boxed{2}, 1, 5, \boxed{7}, 17, 31, \boxed{65}], 127, 257, 511].$$

$$x_2'' = [2, 1, \boxed{5}, 7, 17, \boxed{31}, 65, 127, \boxed{257}], 511].$$

Exercise: Parameters r, l, k

Exercise 1. Consider $z = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$ with parameters $r = 5$, $l = 2$, $k = 2$. Mark all windows and the sampled elements inside each window, following the notation of the examples.

Exercise 2. The following windows have been identified on $y = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]$:

$$y_1 = [\boxed{10}, 20, 30, \boxed{40}, 50, 60, \boxed{70}], 80, 90, 100].$$

$$y_2 = [10, 20, 30, \boxed{40}, 50, 60, \boxed{70}, 80, 90, \boxed{100}].$$

Determine the parameters r , l , and k .

Exercise: Parameters r, l, k – Solutions

Exercise 1. With $r = 5$, $l = 2$, $k = 2$: sampling step $= \frac{r-1}{2l-2} = 2$, so $2l - 1 = 3$ elements are sampled per window.

$$z_1 = [\boxed{1}, 2, \boxed{3}, 4, \boxed{5}, 6, 7, 8, 9, 10].$$

$$z_2 = [1, 2, \boxed{3}, 4, \boxed{5}, 6, \boxed{7}, 8, 9, 10].$$

$$z_3 = [1, 2, 3, 4, \boxed{5}, 6, \boxed{7}, 8, \boxed{9}, 10].$$

Exercise 2.

- ▶ Each window spans 7 elements $\Rightarrow r = 7$.
- ▶ Each window contains $2l - 1 = 3$ sampled elements $\Rightarrow l = 2$.
- ▶ Windows start 3 positions apart $\Rightarrow k = 3$.

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Anomaly Detection

Anomaly Detection as a Special Case of Classification

Anomaly detection is classification with **one class**: the normal class.

- ▶ Laws are extracted from **normal** data only.
- ▶ Instead of one scalar per class, we compute one score **per embedded window**, giving a score *vector*.
- ▶ **Peaks** in the score vector mark the anomaly locations.

Step 1: Laws from Normal Data

Normal data: $a = [1, 1, 2, 3, 5, 8, 13]$, parameters
 $r = 3$, $l = 2$, $k = 1$.

$$L = \begin{bmatrix} -0.8507 & -0.8507 & -0.8507 & -0.8507 & -0.8507 \\ 0.5257 & 0.5257 & 0.5257 & 0.5257 & 0.5257 \end{bmatrix}. \quad (12)$$

All five laws are identical because every window follows the same Fibonacci rule.

Step 2: Embedding the Test Instance

Test instance (anomalies at positions 0 and 3):

$x = [6, 1, 2, 8, 5, 8, 13]$.

$$E = \begin{bmatrix} 6 & 1 \\ 1 & 2 \\ 2 & 8 \\ 8 & 5 \\ 5 & 8 \\ 8 & 13 \end{bmatrix}. \quad (13)$$

Row j is the pair (x_j, x_{j+1}) .

Step 3: Projection

$$F = E \cdot L = \begin{bmatrix} -0.753 & -0.753 & -0.753 & -0.753 & -0.753 \\ 0.090 & 0.090 & 0.090 & 0.090 & 0.090 \\ 0.304 & 0.304 & 0.304 & 0.304 & 0.304 \\ -0.443 & -0.443 & -0.443 & -0.443 & -0.443 \\ -0.005 & -0.005 & -0.005 & -0.005 & -0.005 \\ 0.002 & 0.002 & 0.002 & 0.002 & 0.002 \end{bmatrix}. \quad (14)$$

Rows 0 and 3 have the largest absolute values.

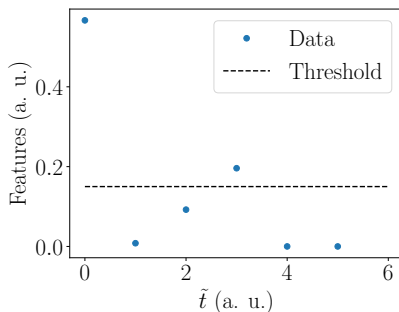
Step 4: Anomaly Scores

Average F^2 over the laws (columns), keeping one score per row:

$$s_j = \text{mean}_m(F_{jm}^2). \quad (15)$$

$$s = [\underbrace{0.567}_{\text{anomaly}}, 0.008, 0.092, \underbrace{0.196}_{\text{anomaly}}, 0.000, 0.000]. \quad (16)$$

Threshold $\tau = 0.15$: gives anomalies (peaks) at 0 and 3.



The mean can be replaced by a **percentile** (e.g. 5th) for robustness.

Index Mapping: Score Index to Original Position

The score vector has $N - l + 1$ elements. Score s_j is computed from window

$$E_j = [x_j, x_{j+1}, \dots, x_{j+l-1}], \quad (17)$$

so a peak at index j indicates an anomaly at original position j .
Notebook example ($l = 2$):

- ▶ $s_0 = 0.567 > \tau \Rightarrow$ anomaly at position 0. ✓
- ▶ $s_3 = 0.196 > \tau \Rightarrow$ anomaly at position 3. ✓

Exercise: Index Mapping

$altx$ is applied to a time series of length $N = 10$ with $l = 3$.

Part (a). How many elements does the anomaly score vector s have?

Part (b). The score vector has peaks at indices $j = 1$ and $j = 6$. What are the corresponding anomaly positions in x ?

Exercise: Index Mapping – Solutions

Part (a).

$$N - l + 1 = 10 - 3 + 1 = 8. \quad (18)$$

Part (b). A peak at index j indicates an anomaly at position j :

- ▶ Peak at $j = 1 \Rightarrow$ anomaly at position 1.
- ▶ Peak at $j = 6 \Rightarrow$ anomaly at position 6.

Summary of ALT Method

- ▶ Extract subsequences and embed into symmetric matrices.
- ▶ Compute eigenvectors of smallest absolute eigenvalues — these are the laws.
- ▶ Multiply embedded test instances with class-specific laws.
- ▶ Compute summary statistics (e.g., mean of squared values).
- ▶ Classify by choosing the class that minimizes the statistic.
- ▶ By only taking statistic over the laws, the peaks represent anomalies.